AN IMPROVED UPPER BOUND FOR BELLMAN'S LOST-IN-A-FOREST PROBLEM IN ISOSCELES TRIANGLES

ALEXANDER TEMEREV

Abstract. We prove a new upper bound for the escape length of isosceles triangles in Bellman's lost-in-a-forest problem, on an entire interval of base angles inside the unsolved range. We construct an explicit piecewise-polynomial family of three-segment paths $S(u)$, $u = \tan(\beta/2)$, with isosceles-trapezoidal convex hull ("staples"), and prove that $S(u)$ escapes the triangle $T(\beta)$ with unit legs and base angle β, and is strictly shorter than every escaping square path, for all $\beta \in [31.08^\circ, 42.16^\circ]$; on $[31.50^\circ, 42.16^\circ]$ it is also strictly shorter than the diameter, the scaled Zalgaller caliper, and the Besicovitch–Movshovich zee. This interval strictly contains $(32.36^\circ, 41.34^\circ)$, the range on which Ward's square path was the shortest known candidate, so the square path is removed from the frontier everywhere, and the trapezoidal improvement Ward reported numerically in 2008 becomes a theorem, with the enlarged range he predicted certified at both ends. The proofs eliminate the continuum of positions and headings exactly: escape from a triangle reduces, via the planar case of a containment theorem of Lutwak, to the containment of a disk in an explicit Minkowski sum, and every resulting inequality is a rational-polynomial sign condition verified by exact Sturm certificates. At the type example, the golden gnomon (apex 108°), the bound reads $\ell(T) \leq \frac{3751558+2\sqrt{20693661817348}}{10^7} = 1.2849616\ldots$ against Ward's $1.2934740\ldots$, with the closed-form bracket $\frac{3}{10}\sqrt{25-5\sqrt{5}} \leq \ell(T)$. Optimality is not claimed.

1. Introduction

Bellman [2] asked for the shortest walk guaranteed to leave a forest of known shape when the hiker knows neither their position nor their heading. Formally, a rectifiable path $P \subset \mathbb{R}^2$ is an *escape path* for a compact convex region T if no congruent copy of P lies in $\operatorname{int} T$; the *escape length* $\ell(T)$ is the infimum of the lengths of such paths [8].

Seventy years of improvements. Progress has come one shape — and one mechanism — at a time. Gross [11] settled the disk: the diameter walk is optimal. Isbell [12], completed by Joris [13], settled the half-plane at known distance. Zalgaller [19] settled the infinite strip with the *caliper*, of length $\zeta = 2.2782916\ldots$ per unit width (see also [1, 4]). Gerriets and Poole [9] proved that all *fat* regions — those containing a 60° rhombus on a diameter — are escaped optimally by a diameter segment. Besicovitch's zigzag [3] for the equilateral triangle was proven optimal by Coulton and Movshovich [5], and Movshovich [15] extended the optimality of the associated *zee* zigzags to all isosceles triangles with base angles in $[45^\circ, 60^\circ]$ (see also [16]).

Why triangles. The solved cases sort into two one-dimensional mechanisms. Fat regions are governed by the *diameter*: the longest chord already escapes optimally. Lean regions are governed by *width*: an arc of width w cannot be hidden in a region of width w, so the scaled caliper gives $\ell(K) \leq \zeta\, w(K)$, exact for the strip and asymptotically right for long thin regions. Triangles belong to neither regime: a triangle is never fat (its diameter is one of its sides,

Date: July 21, 2026.

The computations, the verification scripts, and the initial draft of this manuscript were produced with the assistance of Claude Fable 5 (Anthropic). Verification code is provided as ancillary files.

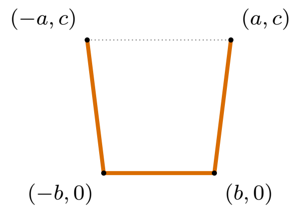

Figure 1. A staple and its isosceles-trapezoidal hull: the bar from $(-b, 0)$ to $(b, 0)$, two equal legs to $(\pm a, c)$, and the dotted top edge completing the hull.

and the rhombus needs room on both sides of that chord), and it is lean only in the limit of small base angle. They are thus the simplest forests squeezed strictly between the two solved mechanisms, and each solved triangular case has required a genuinely new path: the zigzags of Besicovitch and Movshovich on $[45°, 60°]$. Below 45° the problem is open, and the isosceles family $T(\beta)$ — unit legs, base angles β, base $2\cos\beta$ — is its standard laboratory.

Ward [17] explored this open range numerically. He compared the scaled caliper, the zee, the diameter, and a fourth candidate, the *square path* (three sides of a square), and found the square path shortest among the four for $32.36° < \beta < 41.34°$; from this he concluded (his Proposition 15) that "there exists at least one more type of solution for bounded forests." He then recorded the following, which we quote in full since it is the subject of this paper:

> "Some numerical experiments indicate that the results for the square can be improved slightly by increasing the lower base to form a path with an isosceles trapezoidal convex hull. This leads to shorter path lengths and a slightly larger range over which the path surpasses Zalgaller and Besicovitch." [17, p. 49]

He published no coordinates or lengths for this improvement. Gibbs [10], using an uncertified Monte-Carlo search, later placed a "trapezium" phase near base angles 38°–42°. Recent work of Deng [6, 7] formulates the general problem as a boundary-intersection problem and studies finite traveling-salesman discretizations. Our aim is complementary: for triangular forests the continuum of positions and headings can be eliminated *exactly* — a path escapes if and only if one explicit Minkowski sum contains a disk (Section 2) — and candidate paths then admit finite, checkable certificates. No sampled positions or orientations occur in any proof below.

Main result. A *staple* is a three-segment path whose four vertices $(\pm b, 0), (\pm a, c)$ form an isosceles trapezoid: a bar of length $2b$ with two equal legs attached at equal obtuse angles (Figure 1). Ward's square path is the case $a = b$, $c = 2b$; releasing the right angles is what saves length. The name is borrowed from Movshovich and Wetzel: their *staples* [16, §3.2], after [5], are the three-segment arcs whose endpoints lie on the base of the triangle and whose two middle vertices lie on the two arms — exactly the contact pattern of our path at a critical placement (Figure 3) — and their Lemma 9 shows that in the zee range, base angles above 45°, every staple is too long to be optimal. Theorem 1 certifies that below 45° the staples take the lead.

Theorem 1. *Let $u = \tan(\beta/2)$, and let $S(u)$ be the staple whose parameters $a(u), b(u), c(u)$ are the explicit rational polynomials of Section 3 (degrees at most 6; 51 rational coefficients in total, printed in Appendix A and listed in the ancillary file* `interval_construction.json`*). Then for every $u \in [0.278, 0.3855]$, that is for every base angle $\beta = 2\arctan u \in [31.0719°, 42.1633°]$:*

(i) *$S(u)$ is an escape path for $T(\beta)$;*
(ii) *$|S(u)| < 3s$ for every square path of side s that escapes $T(\beta)$ — in particular $|S(u)|$ is strictly smaller than Ward's square-path length in both of its regimes;*

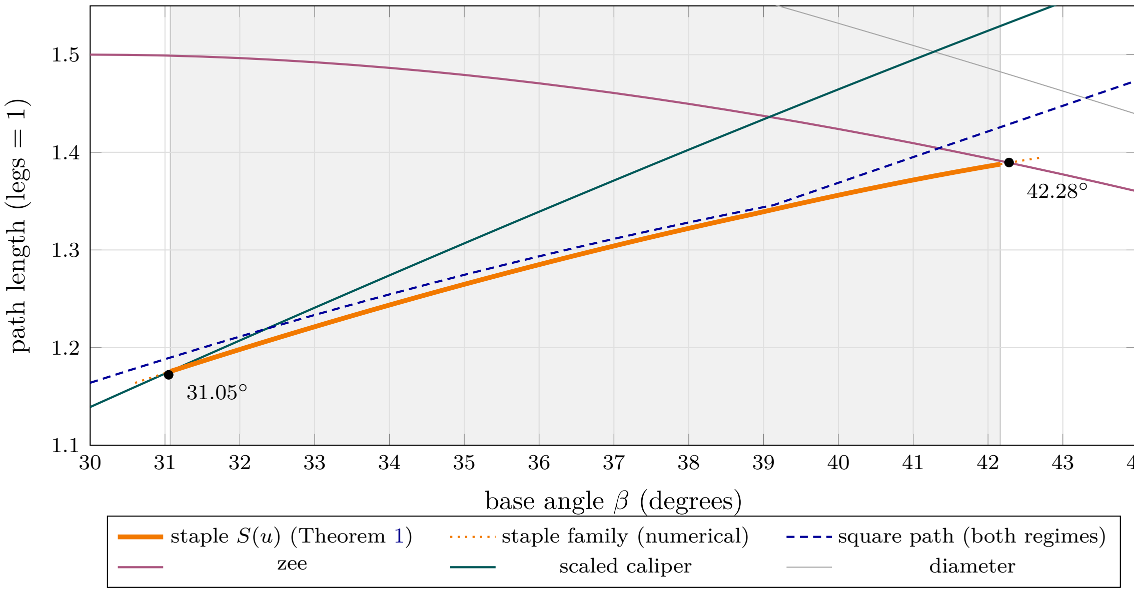

FIGURE 2. The five candidate lengths against the base angle β. The shaded band $[31.07°, 42.16°]$ is the certified domain of Theorem 1: there the staple $S(u)$ (thick) escapes and lies strictly below the square-path curve, whose two regimes meet in the visible kink at $\beta_0 = 39.132°$. On $[31.50°, 42.16°]$ it lies below all four competitors (Corollary 2). Dotted continuations and the two crossing points at 31.05° and 42.28° are numerical (Section 5); everything else in the band is proved.

(iii) $|S(u)|$ *is strictly smaller than the critical zee length* $\frac{6\sin\beta\cos\beta}{\sqrt{1+8\sin^2\beta}}$ *(Ward's closed form* [17]*, rescaled to legs* 1*; the arcs are the* z_β *of* [16, 15]*) and the diameter* $2\cos\beta$*; and for* $u \geq 0.282$ *(i.e.* $\beta \geq 31.4969°$*) also strictly smaller than the caliper length* $\zeta \sin\beta$*.*

Corollary 2 (Ward's enlargement, certified)**.** *For every base angle* $\beta \in [31.50°, 42.16°] \supsetneq (32.36°, 41.34°)$*, the staple* $S(u(\beta))$ *is an escape path for* $T(\beta)$ *strictly shorter than each of the four candidates compared by Ward: the diameter, the scaled Zalgaller caliper, the Besicovitch–Movshovich zee, and every escaping square path.*

Figure 2 shows the five length curves. Two features are the content of Theorem 1: the staple curve lies strictly below the square-path curve across a band that covers Ward's entire square-path range, so the square path is now nowhere the best known candidate; and it lies below all four competitors on $[31.50°, 42.16°]$, certifying the enlargement Ward predicted — at both ends.

The type example. At a single pleasant triangle the bound becomes a closed-form statement. The *golden gnomon* is the isosceles triangle with unit legs, apex 108° (base angles 36°), and base $\varphi = (1 + \sqrt{5})/2$.

Theorem 3. *Let* T *be the golden gnomon, and let* S *be the polygonal path through*

$$\left(-\frac{2430971}{10^7}, \frac{4515022}{10^7}\right), \quad \left(-\frac{1875779}{10^7}, 0\right), \quad \left(\frac{1875779}{10^7}, 0\right), \quad \left(\frac{2430971}{10^7}, \frac{4515022}{10^7}\right).$$

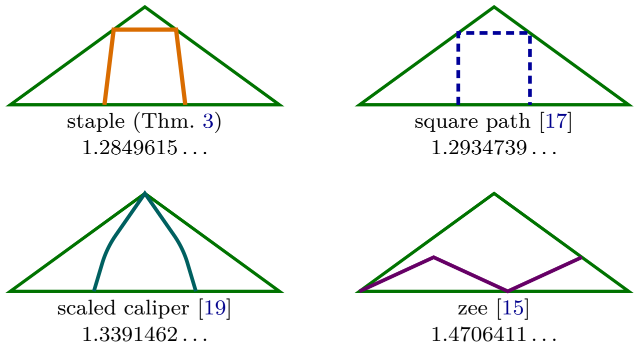

FIGURE 3. The four candidates at the golden gnomon (legs 1, apex $108°$, diameter 1.6180340), each drawn at a critical placement: each path's hull fits T only in touching position. The staple, square path, and caliper all touch the base with their endpoints; the caliper's peak sits exactly at the apex. The staple of Theorem 3 achieves criticality at the smallest length.

Then S escapes T, and

$$\ell(T) \leq |S| = \frac{3751558 + 2\sqrt{20693661817348}}{10^7} = 1.28496153349145\ldots < 1.284962.$$

In particular $|S|$ is smaller than Ward's square-path length $1.293473869\ldots$, *by* 8.512×10^{-3}.

Corollary 4 (a rigorous bracket)**.** *For the golden gnomon,*

$$\frac{3}{10}\sqrt{25 - 5\sqrt{5}} \;=\; 1.115244103\ldots \;\leq\; \ell(T) \;<\; 1.284962.$$

Figure 3 compares the path of Theorem 3 with the four standard candidates at the gnomon. Sections 2–4 contain the proofs; Section 5 separates out the numerical evidence (stationary staples, family crossings) that guided the constructions but enters no proof. We emphasize what is *not* claimed: optimality of the staple, within its family or otherwise. All code and data accompanying the paper are available at `https://github.com/atemerev/bellman-staple` and as arXiv ancillary files.

2. The escape criterion

Throughout, $h_C(u) = \sup_{x\in C}\langle x, u\rangle$ denotes the support function of a compact convex set C, and R_θ the rotation of $\mathbb{R}^2$ by angle θ. We use two standard facts: $h_{C\oplus D} = h_C + h_D$ for the Minkowski sum, and $h_{RC}(u) = h_C(R^{-1}u)$.

Lemma 5 (fit criterion for triangles)**.** *Let $T = \{x : \langle n_i, x\rangle \leq t_i,\ i = 1, 2, 3\}$ be a triangle with unit outward normals n_i and side lengths s_i, so that $\sum_i s_i n_i = 0$ and $\sum_i s_i t_i = 2\,\mathrm{Area}(T)$. For a compact convex C there exists a translation t with $C + t \subseteq T$ if and only if*

$$s_1 h_C(n_1) + s_2 h_C(n_2) + s_3 h_C(n_3) \;\leq\; 2\,\mathrm{Area}(T).$$

Since $\sum_i s_i\, h_C(n_i) = 2V(C, T)$, twice the mixed area, the criterion reads: *C translates into T if and only if* $V(C, T) \leq V(T, T) = \mathrm{Area}(T)$. In this mixed-volume form (for simplices in all dimensions) the criterion is a result of Lutwak [14, Prop., p. 230 and the remark following Cor. 3], with ancestry in Weil's characterization of translative containment [18]. We include the short linear-programming proof both to keep the note self-contained and because its sufficiency half produces the explicit feasible translations used in our certificates. We have not found this finite disk-containment specialization in the escape-path literature.

Proof. $C + t \subseteq T$ is equivalent to the linear system $\langle n_i, t\rangle \le c_i := t_i - h_C(n_i)$, $i = 1, 2, 3$.

Necessity. If t solves the system then $0 = \langle \sum_i s_i n_i, t\rangle \le \sum_i s_i c_i$, which rearranges to the stated inequality.

Sufficiency. Suppose $\lambda := \big(\sum_i s_i c_i\big)/\big(2\,\mathrm{Area}(T)\big) \ge 0$. The affine system $\langle n_i, t_0\rangle = c_i - \lambda t_i$ ($i = 1, 2, 3$; three equations in two unknowns) is solvable: the unique linear dependency among the rows n_i has coefficients s_i, and $\sum_i s_i(c_i - \lambda t_i) = \sum_i s_i c_i - \lambda \cdot 2\,\mathrm{Area}(T) = 0$ is precisely its compatibility condition. For this t_0,

$$\{t : \langle n_i, t\rangle \le c_i\} \;=\; \lambda T + t_0 \;\ne\; \emptyset. \qquad \square$$

Note the geometric content of sufficiency: *the set of admissible translations is itself a homothet of the triangle*, of ratio λ.

Proposition 6 (escape criterion)**.** *With T, $n_i = (\cos\alpha_i, \sin\alpha_i)$, s_i as above, a path P with $H = \mathrm{conv}(P)$ is an escape path for T if and only if the Minkowski sum*

$$W \;:=\; s_1\, R_{-\alpha_1} H \;\oplus\; s_2\, R_{-\alpha_2} H \;\oplus\; s_3\, R_{-\alpha_3} H$$

contains the closed disk of radius $2\,\mathrm{Area}(T)$ centered at the origin. If H is a polygon, then so is W.

Proof. A placed congruent copy of P lies in the open convex set $\mathrm{int}\, T$ if and only if its convex hull does, and rigid motions factor as a rotation followed by a translation; if a placed copy is not contained in the interior, the connected path, starting inside, must meet ∂T — the hiker escapes (cf. [8]). Thus P escapes iff for every θ no translate of $R_\theta H$ lies in $\mathrm{int}\, T$; by Lemma 5 this holds iff for all θ, $\sum_i s_i\, h_{R_\theta H}(n_i) \ge 2\,\mathrm{Area}(T)$ (the closed inequality: a hull that can only be translated to touch ∂T still reaches the boundary). Writing $h_{R_\theta H}(n_i) = h_H(R_{-\theta} u(\alpha_i)) = h_{R_{-\alpha_i} H}(u(-\theta))$ with $u(\psi) = (\cos\psi, \sin\psi)$ and summing, $\sum_i s_i h_{R_\theta H}(n_i) = h_W(u(-\theta))$. The condition $h_W(u) \ge 2\,\mathrm{Area}(T)$ for every unit u is equivalent to $W \supseteq \overline{D}(0, 2\,\mathrm{Area}(T))$. $\square$

Remark 7. Two computational forms of the inclusion are used below. For a fixed triangle, W is an explicit polygon and the inclusion is a list of edge-distance inequalities (Section 4 and Figure 6). For a one-parameter family it is more convenient to avoid tracking the boundary of W: a convex polygon is the intersection of its edge half-planes, and every edge normal of a Minkowski sum is an edge normal of one of the summands, so it suffices to verify $h_W(\nu) \ge 2\,\mathrm{Area}(T)\cdot|\nu|$ for the twelve candidate directions $\nu = R_{-\alpha_i} g$, with g ranging over the four edge normals of H — and $h_W(\nu)$ may be replaced by any convenient lower bound, e.g. by evaluating each summand's support at a fixed vertex (Section 3).

Remark 8. For the unit equilateral triangle the criterion reads: a path escapes iff the Minkowski sum of three copies of its hull rotated by $0°, 120°, 240°$ contains the disk of radius $2\,\mathrm{Area} = \frac{\sqrt{3}}{2}$. Besicovitch's zigzag [3, 5] is exactly critical in this sense: its Minkowski sum circumscribes that disk.

3. Proof of Theorem 1

Parametrize by $u = \tan(\beta/2)$, so that $\sin\beta = \frac{2u}{1+u^2}$ and $\cos\beta = \frac{1-u^2}{1+u^2}$ are rational functions of u. All triangle data — normals, side lengths, the rotations of Proposition 6, the disk radius $2\,\mathrm{Area} = \sin 2\beta$ — are then rational in u, and for a staple whose parameters a, b, c are *polynomials* in u with rational coefficients, every certificate of Remark 7, after clearing the (positive) denominators and squaring away the (sign-certified) square roots, is the strict

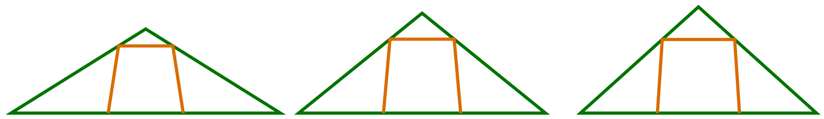

FIGURE 4. The certified family $S(u)$ inside $T(\beta)$ at $\beta = 32°$, $39°$, $42°$, each at a critical placement (the hull fits only in touching position).

positivity of a polynomial in $\mathbb{Q}[u]$ on a rational interval — decidable by Sturm's theorem in exact integer arithmetic.

3.1. **The construction.** The ancillary file `interval_construction.json` defines three polynomial triples $(a_i, b_i, c_i) \in \mathbb{Q}[u]^3$, of degrees 6, 2, and 6 on the respective pieces

$$[0.278,\ 0.3530], \qquad [0.3530,\ 0.353399], \qquad [0.353399,\ 0.3855],$$

and $S(u)$ is the path through $(-a_i(u), c_i(u))$, $(-b_i(u), 0)$, $(b_i(u), 0)$, $(a_i(u), c_i(u))$. All 51 coefficients are rational numbers, printed in full in Appendix A; the machine-readable copy is read afresh by the verification script, which certifies $a_i, b_i, c_i > 0$ together with every inequality used below. Figure 4 shows the family at three angles. The middle piece is a deliberately thin bridge: near $u \approx 0.3534$ the numerically optimal staple changes which edges of its Minkowski sum touch the disk (Section 5), and the two outer fits track the two contact regimes; the quadratic bridge hands over between them, and its exact endpoints are immaterial — every certificate is checked on each piece separately.

3.2. **Eliminating square paths.**

Lemma 9 (square paths from inscribed squares)**.** *Fix β and suppose some square Q_0 of side s_0 satisfies $Q_0 \subseteq T(\beta)$. Then every square path of side $s < s_0$ fails to escape $T(\beta)$; consequently every escaping square path has length $\geq 3s_0$.*

Proof. The hull of a square path of side s is the full square. Shrinking Q_0 about its center by the factor $s/s_0 < 1$ places a congruent copy of that hull in $\operatorname{int} Q_0 \subseteq \operatorname{int} T(\beta)$. $\square$

The two inscribed squares we use are Ward's, in exact form (Figure 5): the base-aligned square of side

$$s_A = \frac{2 \sin\beta \cos\beta}{\sin\beta + 2\cos\beta},$$

with both bottom corners on the base and both top corners on the legs, and the leg-aligned square of side

$$s_B = \frac{\sin\beta}{\sin\beta + \cos\beta},$$

with one full side on a leg, its far corner *at the apex*, and one corner on the base. The stated incidences are polynomial identities in u; the remaining vertex–halfplane constraints are certified by Sturm checks. Ward's two square-path formulas $3s_A$ and $3s_B$ exchange at the root $\beta_0 = 39.1320\ldots°$ of $\sin\beta\,(2\cos\beta - 1) = 2\cos\beta\,(1 - \cos\beta)$ (his "approximately 39.1°"), the kink visible in Figure 2.

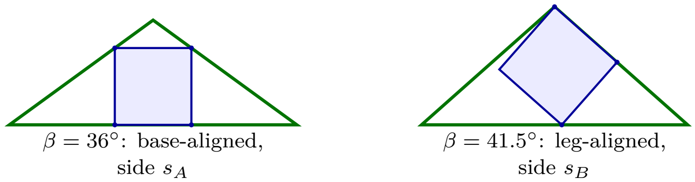

FIGURE 5. The two inscribed squares behind Lemma 9. Left: the base-aligned square, with both bottom corners on the base and both top corners on the legs. Right: the leg-aligned square, with one full side on a leg, its far corner at the apex, and one corner on the base (contacts marked). Any escaping square path must have side at least $\max(s_A, s_B)$; the certificates prove $|S(u)| < 3s_A$ for $u \le 0.365$ and $|S(u)| < 3s_B$ for $u \ge 0.36$, which together cover the whole interval.

3.3. **The certificates.** Every assertion of Theorem 1 is now a strict polynomial inequality on a rational interval. (i): the twelve support-dominance inequalities of Remark 7 per piece, with fixed support-test vertices (their values are lower bounds for h_W); for the four leg-normal directions the inequality is squared once, with h-positivity certified separately. (ii): by Lemma 9 it suffices that $|S(u)| < 3s_A(u)$ for $u \le 0.365$ and $|S(u)| < 3s_B(u)$ for $u \ge 0.36$, once the two squares are certified inscribed; each comparison squares away the single root in $|S(u)|$. (iii): the diameter and caliper comparisons are of the same shape, using the outward-rounded bracket $\zeta > 2.2782916$ from the published closed form; the zee comparison carries one further square root and is squared twice, with both intermediate signs certified. (The zee length in (iii) is Ward's closed form Q_B [17] rescaled from unit base to unit legs.)

In total 116 rational-polynomial certificates (degrees ≤ 40) are verified by endpoint signs and exact Sturm root counts. The Zalgaller constant is the only transcendental input and is handled separately by directed evaluation. The verification script is the ancillary file `interval_certificate.py`; it re-derives every polynomial from the construction data and the exact triangle geometry. No floating point enters any accepted certificate. ☐

Corollary 2 follows by intersecting the intervals in Theorem 1(ii)–(iii): all four comparisons hold simultaneously for $u \in [0.282, 0.3855] \supseteq [31.50°, 42.16°]$ in base angle. ☐

4. THE GOLDEN GNOMON: AN EXACT CERTIFICATE AT THE TYPE EXAMPLE

Theorem 3 is the case $\beta = 36°$ with a fixed rational staple, and its certificate is short enough to display in full.

4.1. **Exact data.** Place T with base from $(-\varphi/2, 0)$ to $(\varphi/2, 0)$ and apex $(0, \sin 36°)$; the legs have length 1 since $\cos 36° = \varphi/2$. The sides have lengths $(s_1, s_2, s_3) = (\varphi, 1, 1)$ and unit outward normals

$$n_1 = (0, -1) = u(-90°), \qquad n_{2,3} = (\pm \sin 36°,\ \cos 36°) = u(54°),\, u(126°),$$

and one checks exactly that $\varphi\, n_1 + n_2 + n_3 = 0$. Two golden identities drive all the arithmetic:

$$2\,\mathrm{Area}(T) = \varphi \sin 36° = \sin 72°, \qquad \big(2\,\mathrm{Area}(T)\big)^2 = \frac{5+\sqrt{5}}{8},$$

with $\sin 36° = \frac{1}{4}\sqrt{10 - 2\sqrt{5}}$. All quantities below live in the real field $\mathbb{Q}\big(\sqrt{5}, \sqrt{10-2\sqrt{5}}\,\big)$. The hull H of the staple is the isosceles trapezoid with vertices $(\pm b, 0)$ and $(\pm a, c)$, where

$$a = \frac{2430971}{10^7}, \qquad b = \frac{1875779}{10^7}, \qquad c = \frac{4515022}{10^7},$$

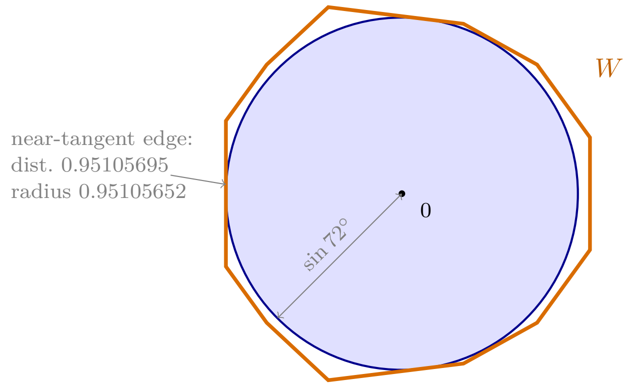

Figure 6. The proof of Theorem 3 in one picture: the Minkowski sum $W = \varphi R_{90^\circ} H \oplus R_{-54^\circ} H \oplus R_{-126^\circ} H$ contains the closed disk of radius $2\,\mathrm{Area}(T) = \sin 72^\circ$. Every edge line of W keeps distance $\geq \sin 72^\circ$ from the origin; by Proposition 6 the staple escapes. Three of the twelve edges are tangent to the disk to within 5×10^{-7}.

and $|S| = 2b + 2\sqrt{(a-b)^2 + c^2}$ as stated.

4.2. **The Minkowski sum.** By Proposition 6 with rotation matrices

$$R_{-\alpha_1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_{-\alpha_{2,3}} = \begin{pmatrix} \pm\sin 36^\circ & \cos 36^\circ \\ -\cos 36^\circ & \pm\sin 36^\circ \end{pmatrix},$$

the path S escapes T iff $W = \varphi R_{-\alpha_1} H \oplus R_{-\alpha_2} H \oplus R_{-\alpha_3} H$ contains $\overline{D}(0, \sin 72^\circ)$. The vertices of W are sums of one vertex from each summand; W is the twelve-edge polygon of Figure 6, symmetric about the horizontal axis (the axial symmetry of H conjugates the two leg copies and fixes the base copy).

4.3. **The twelve inequalities.** For a convex polygon given by a positively oriented vertex cycle $w_1, \ldots, w_{12}$, the inclusion $\overline{D}(0, \rho) \subseteq W$ is equivalent to: (i) all consecutive cross products $(w_{k+1} - w_k) \times (w_{k+2} - w_{k+1}) > 0$ (convexity), (ii) $w_k \times w_{k+1} > 0$ for all k (the origin lies inside), and (iii) for every edge,

$$\big(w_k \times w_{k+1}\big)^2 \ \geq\ \rho^2\, |w_{k+1} - w_k|^2,$$

i.e., every edge line keeps distance at least ρ from the origin. Here $\rho^2 = \frac{5+\sqrt{5}}{8}$ and all w_k are explicit vectors with entries in $\mathbb{Q}(\sqrt{5}, \sqrt{10 - 2\sqrt{5}}\,)$, so each of (i)–(iii) is the sign of an element of that field. Table 1 lists the distances and the certificate values $D_k = (w_k \times w_{k+1})^2 - \rho^2 |w_{k+1} - w_k|^2$; by the axial symmetry of W only seven are distinct. All are strictly positive, with minimum margin $D = 5.13 \times 10^{-7}$.

The ancillary file `tight_staple_proof.py` constructs these field elements symbolically and asks SymPy for their exact algebraic signs; decimal evaluation is used only for the displayed values. It also verifies exactly that all 64 candidate vertex sums lie on or inside every edge half-plane, so the cycle of Table 1 is indeed the boundary of W. Thus the disk is contained in W, and Proposition 6 proves that S escapes T.

edge of W	multiplicity	distance to 0	certificate D_k
left vertical	1	0.951056952	$+5.131 \times 10^{-7}$
$(-0.951, -0.393) \to (-0.731, -0.697)$	2	1.000620318	$+1.361 \times 10^{-2}$
$(-0.731, -0.697) \to (-0.398, -1.007)$	2	1.007883096	$+2.304 \times 10^{-2}$
$(-0.398, -1.007) \to (0.333, -0.917)$	2	0.951057083	$+5.837 \times 10^{-7}$
$(0.333, -0.917) \to (0.731, -0.697)$	2	0.963597925	$+4.969 \times 10^{-3}$
$(0.731, -0.697) \to (1.016, -0.304)$	2	1.000620318	$+2.287 \times 10^{-2}$
right vertical	1	1.016323686	$+4.731 \times 10^{-2}$
		disk radius $\sin 72°$	0.9510565

TABLE 1. The distance certificates for the twelve edges of W (mirror-symmetric edges listed once). $D_k > 0$ is the exact inequality (iii); decimals are 60-digit evaluations truncated for display.

4.4. **Length comparisons.** With $t = \tan 36° = \sqrt{10 - 2\sqrt{5}}/(1 + \sqrt{5})$:

staple (Thm. 3):	$\dfrac{3751558 + 2\sqrt{20693661817348}}{10^7}$	$= 1.2849615\ldots$	
square path [17]:	$\dfrac{3\varphi t}{t+2}$	$= 1.2934739\ldots$	$(+8.512 \times 10^{-3})$
scaled caliper [19]:	$\zeta \sin 36°$	$= 1.3391462\ldots$	$(+5.418 \times 10^{-2})$
zee [15]:	$\dfrac{3\varphi\sqrt{5}}{\sqrt{50 + 2\sqrt{5}}}$	$= 1.4706411\ldots$	$(+1.857 \times 10^{-1})$
diameter:	φ	$= 1.6180340\ldots$	$(+3.331 \times 10^{-1})$

All five are explicit algebraic numbers except ζ, for which the directed evaluation $\zeta > 2.2782916$ of its published closed form suffices. This completes the proof of Theorem 3. □

Remark 10. At $\beta = 36°$ the base-aligned construction gives the valid square path in the table. The leg-aligned value $3s_B = 1.2624233\ldots$ is smaller, but it is not an escape bound there: that square fits strictly inside T (with clearance 6.3×10^{-3}), illustrating why Section 3 uses inscribed squares only as *lower* bounds for escaping square paths.

Proof of Corollary 4. Inside T place the isosceles triangle with base angle 45°, with its base on the base of T and its apex at the apex of T. Its height is $h = \sin 36°$ and its legs have length $\sqrt{2}h$. The escape length of a unit-leg 45° triangle is $3/\sqrt{5}$ by [5, 15]. Monotonicity under inclusion and scaling therefore give

$$\ell(T) \geq \sqrt{2}h\,\frac{3}{\sqrt{5}} = \frac{6}{\sqrt{10}}\sin 36° = \frac{3}{10}\sqrt{25 - 5\sqrt{5}}.$$

The upper bound is Theorem 3. □

5. Numerical discovery and conjectural context

Everything in this section is numerical evidence. It is included to explain how the exact witnesses were found and to identify plausible next questions. It is not used in Theorems 1 or 3. In particular, a stationary point within the staple family is not a proof of optimality, either within that family or among all escape paths.

For a symmetric staple at the gnomon, numerical constrained optimization suggests a Karush–Kuhn–Tucker stationary point at

$$a = 0.2430970\ldots,\quad b = 0.1875777\ldots,\quad c = 0.4515020\ldots,\quad L = 1.2849609182\ldots,$$

at which one self-symmetric edge and one mirror pair of edges of W are tangent to the disk. The rational witness of Theorem 3 is a nearby rational point on the feasible side, only 6.16×10^{-7} longer; the exact disk-containment check, not the stationarity computation, is what makes it a theorem. Along the family, the tangency pattern changes twice near $\beta \approx 38.9°$ — the two interior breakpoints of the construction in Section 3.

Continuation of the same stationary branch suggests crossings with the scaled caliper at $\beta \approx 31.05°$ and with the zee at $\beta \approx 42.28°$ (the two dots in Figure 2); the latter agrees strikingly with Gibbs's Monte-Carlo estimate 42.3° [10]. These are crossings between three specified candidate families, not proven phase boundaries of the unrestricted problem; what is proved is the slightly smaller interval of Corollary 2. The wider numerical search also produces nonpolygonal-looking hybrids between the staple and caliper regimes, at apex angles roughly 110°–118°; these may be closer to Ward's proposed "Zalgalloids", and turning them into exact objects is a natural next problem.

6. Conclusion

The contribution of this paper is a new, exactly certified upper bound for the escape length of isosceles triangles on the interval $[31.07°, 42.16°]$ of base angles: an explicit trapezoidal family that removes Ward's square path from the frontier everywhere and beats all four of his comparison families on $[31.50°, 42.16°]$, together with a finite-certificate method that replaces the continuum of placements by polygon arithmetic. No optimality is claimed; the gap in Corollary 4 remains substantial, and determining whether the stationary staple, a nearby hybrid, or an entirely different path is optimal is open.

Ancillary files.

- `tight_staple_proof.py` verifies Theorem 3: it reconstructs W, determines the exact algebraic signs in Table 1, and checks the length comparisons.
- `interval_construction.json` lists the exact rational coefficients of the staple family $S(u)$.
- `interval_certificate.py` verifies all 116 Sturm certificates of Theorem 1 and Corollary 2.

All three files, together with the numerical discovery code and its data, are maintained at `https://github.com/atemerev/bellman-staple`.

Appendix A. The coefficients of the staple family

Table 2 prints the construction of Theorem 1 in full. The same numbers are listed as integer fractions in the ancillary file `interval_construction.json`, together with the fixed support-test vertex choices used in the escape certificates; the verification script consumes the file, not this table.

References

[1] A. Adhikari and J. Pitman, *The shortest planar arc of width 1*, Amer. Math. Monthly **96** (1989), 309–327.
[2] R. Bellman, *Minimization problem*, Bull. Amer. Math. Soc. **62** (1956), 270.
[3] A. S. Besicovitch, *On arcs that cannot be covered by an open equilateral triangle of side 1*, Math. Gazette **49** (1965), 286–288.

piece 1: $u \in [139/500,\ 353/1000]$			
k	a_k	b_k	c_k
0	0.2180623261593	0.50834630972286	−0.68618491542384
1	−2.98683316746864	−9.67586466442302	16.00857768415038
2	34.80963432476292	88.60308231720582	−120.98302430509386
3	−165.44093496652428	−407.4340115522607	544.85319850303314
4	419.36689348412628	1052.77262491939776	−1412.41888562475624
5	−566.71412345478798	−1463.96980815258684	1965.72502529223156
6	317.71808396659464	860.8859714000805	−1153.82147540598876
piece 2: $u \in [353/1000,\ 353399/1000000]$			
k	a_k	b_k	c_k
0	−44.64121070435796	64.82150688020952	−53.83538739921426
1	258.38271230989518	−370.2997274914329	310.226332252557
2	−371.75125098832224	530.4122974737888	−443.04905105780508
piece 3: $u \in [353399/1000000,\ 771/2000]$			
k	a_k	b_k	c_k
0	598.39329462654	931.9350917140677	−1000.1742396697317
1	−9888.67910670574158	−15411.81001925178432	16547.06303226795252
2	68135.45237045777328	106266.35727563781996	−114104.67388908582498
3	−250502.62369568886828	−391014.4027686377397	419956.42011173402622
4	518368.51791825810486	809875.46729334615888	−870074.61994937309994
5	−572502.14475386179926	−895368.13774997390718	962241.70033218066126
6	263679.85825187420358	412866.98586667984212	−443874.26394488674308

TABLE 2. The 51 coefficients of the staple family: on each piece, $x(u) = \sum_k x_k u^k$ for $x \in \{a, b, c\}$. Every value is an exact rational number whose denominator has the form $2^i 5^j$, printed here as its full terminating decimal.

[4] T. M. Chan, A. Golynski, A. López-Ortiz and C.-G. Quimper, *Curves of width one and the river shore problem*, Proc. 15th Canad. Conf. Comput. Geom. (CCCG), 2003, 73–75.

[5] P. Coulton and Y. Movshovich, *Besicovitch triangles cover unit arcs*, Geom. Dedicata **123** (2006), 79–88.

[6] Z. Deng, *A general solution to Bellman's lost-in-a-forest problem*, arXiv:2412.10686v3 (2025), https://arxiv.org/abs/2412.10686v3.

[7] Z. Deng, *Proof and more variations of Bellman's lost-in-a-forest problem*, arXiv:2606.13987v2 (2026), https://arxiv.org/abs/2606.13987.

[8] S. R. Finch and J. E. Wetzel, *Lost in a forest*, Amer. Math. Monthly **111** (2004), 645–654.

[9] J. Gerriets and G. Poole, *Convex regions which cover arcs of constant length*, Amer. Math. Monthly **81** (1974), 36–41.

[10] P. Gibbs, *Lost in an isosceles triangle*, viXra:1607.0015 (2016).

[11] O. Gross, *A search problem due to Bellman*, RAND Corporation, Research Memorandum RM-1603, 1955.

[12] J. R. Isbell, *An optimal search pattern*, Naval Res. Logist. Quart. **4** (1957), 357–359.

[13] H. Joris, *Le chasseur perdu dans la forêt*, Elem. Math. **35** (1980), 1–14.

[14] E. Lutwak, *Containment and circumscribing simplices*, Discrete Comput. Geom. **19** (1998), 229–235.

[15] Y. Movshovich, *Besicovitch triangles extended*, Geom. Dedicata **159** (2012), 99–107.

[16] Y. Movshovich and J. E. Wetzel, *Escape paths of Besicovitch triangles*, J. Combinatorics **2** (2011), 413–433.

[17] J. W. Ward, *Exploring the Bellman forest problem*, manuscript (2008), http://wardsattic.com/math/BellmanForestProblem/BellmanForestProblem.pdf; archived at https://web.archive.org/web/20250216114838/http://wardsattic.com/math/BellmanForestProblem/BellmanForestProblem.pdf.

[18] W. Weil, *Decomposition of convex bodies*, Mathematika **21** (1974), 19–25.

[19] V. A. Zalgaller, *How to get out of the woods? On a problem of Bellman* (Russian), Matematicheskoe Prosveshchenie **6** (1961), 191–195.

UNIVERSITY OF GENEVA, GENEVA, SWITZERLAND
Email address: alexander.temerev@unige.ch